

Anish kumar M

Agentic AI Engineer | Generative AI Developer | RAG Systems | LLM Application Development

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Professional Summary

Results-driven Agentic AI Engineer with 2+ years of experience in designing, developing, and deploying intelligent AI-powered enterprise applications using CrewAI, LangGraph, LangChain, Flowise, Python, and FastAPI. Experienced in building multi-agent systems, Retrieval-Augmented Generation (RAG) pipelines, AI workflow orchestration, vector database implementations, and enterprise API integrations. Hands-on expertise in prompt engineering, LLM integration, AI agent memory management, Human-in-the-Loop (HITL) workflows, evaluation frameworks, and workflow automation. Proven ability to develop scalable and production-ready Agentic AI solutions for Digital Marketing and Marketing Technology (MarTech) platforms, enabling campaign automation, customer engagement, marketing analytics, and business optimization.

Technical Skills

Agentic AI Frameworks: CrewAI, LangGraph, LangChain, Flowise, AutoGen (Basics), n8n

Programming Languages: Python, SQL, JavaScript (Basics), Bash

Python Libraries: LangChain, LangGraph, CrewAI, FastAPI, Pydantic, Requests, HTTPX, Pandas, NumPy, Sentence Transformers, Transformers, PyTest

LLM Platforms: OpenAI GPT-4o, Claude 3.5 Sonnet, Gemini, Llama 3.1, Mistral, Ollama, LM Studio

Retrieval & Vector Databases: RAG, ChromaDB, FAISS, Pinecone, Hybrid Search, Semantic Chunking

Agent Architecture: Supervisor-Worker, ReAct, Planner-Executor, Sequential Agents, Reflection, Human-in-the-Loop (HITL)

Prompt Engineering: Few-shot Prompting, Chain-of-Thought, Prompt Chaining, Structured Outputs, JSON Mode

Backend & APIs: FastAPI, REST APIs, Async Python, Webhooks, Postman

Workflow Automation: n8n, Power Automate, Batch Processing

Evaluation & Testing: RAGAS, Prompt Evaluation, PyTest, Human Evaluation, Latency Testing

DevOps & Tools: Docker, Git, GitHub Actions, Linux, VS Code, Jupyter Notebook

Professional Experience

Agentic AI Engineer June 2024 – Present

HCL Tech | Chennai, India

Multi-Agent Workflow Development

Designed and developed intelligent multi-agent workflows using CrewAI, LangGraph, and LangChain to automate digital marketing and customer engagement processes. Built specialized AI agents for campaign planning, SEO content generation, lead qualification, customer segmentation, marketing analytics, customer journey mapping, and performance reporting. Developed modular AI workflows capable of coordinating multiple marketing operations while improving productivity, reducing manual effort, and enabling faster campaign execution.

AI Tool Integration & Python Development

Developed reusable Python tools enabling AI agents to securely communicate with CRM platforms, marketing automation systems, analytics tools, REST APIs, and enterprise databases. Built FastAPI-based microservices exposing

AI capabilities through secure REST endpoints while implementing asynchronous processing, request validation, authentication, and exception handling. Integrated AI agents with customer engagement platforms, campaign management systems, and analytics dashboards to automate business workflows.

Retrieval-Augmented Generation (RAG)

Designed and implemented Retrieval-Augmented Generation (RAG) pipelines using LangChain, ChromaDB, and FAISS to enable intelligent retrieval of marketing playbooks, SEO guidelines, campaign documentation, customer personas, and product information. Developed document ingestion pipelines supporting PDF, DOCX, CSV, and JSON files while implementing semantic chunking, metadata filtering, embedding generation, and vector indexing to improve retrieval quality and reduce hallucinations.

LLM Integration & Prompt Engineering

Integrated OpenAI GPT-4o, Claude, Gemini, Ollama, and Llama models into enterprise AI workflows for campaign planning, marketing content generation, customer communication, and business automation. Designed reusable prompt templates using Few-shot Prompting, Chain-of-Thought reasoning, structured JSON outputs, and prompt chaining techniques to improve content consistency, campaign recommendations, AI reasoning, and tool execution accuracy.

Agent Memory & Workflow Governance

Implemented conversation memory using ChromaDB to enable personalized customer interactions and context-aware marketing assistants. Developed Human-in-the-Loop (HITL) workflows for campaign approvals, marketing content validation, customer communication review, and workflow monitoring. Collaborated with cross-functional teams to improve AI reliability, operational transparency, and production readiness.

Evaluation, Deployment & DevOps

Built automated evaluation frameworks to measure AI response quality, retrieval accuracy, campaign recommendation relevance, and latency using RAGAS and custom evaluation scripts. Containerized AI applications using Docker, managed source code with GitHub, and supported CI/CD deployments using GitHub Actions. Assisted in monitoring production AI services and ensuring reliable deployment of enterprise Agentic AI solutions.

Flagship Project 1: AI Marketing Automation Platform

Client: Leading Digital Marketing Solutions Provider

Domain: Digital Marketing

Duration: June 2024 – June 2025

Role: Agentic AI Engineer

Business Challenge

Marketing teams manually planned campaigns, generated SEO content, analyzed advertising performance, qualified leads, and prepared campaign reports across multiple platforms. The lack of automation increased campaign execution time, delayed decision-making, and reduced marketing efficiency.

Architecture & Technical Implementation

Developed an intelligent multi-agent marketing platform using CrewAI, LangGraph, and LangChain to automate digital marketing operations. Built specialized AI agents for Campaign Planning, SEO Content Generation, Ad Performance Analysis, Lead Scoring, Email Content Generation, Customer Segmentation, and Marketing Report Generation. Implemented Retrieval-Augmented Generation (RAG) pipelines over SEO guidelines, campaign documentation, marketing strategies, customer personas, and product catalogs using ChromaDB and FAISS. Integrated AI agents with CRM platforms, Google Analytics, Meta Ads APIs, email marketing systems, and reporting dashboards using FastAPI while implementing conversation memory and workflow automation.

Tech Stack: Python, CrewAI, LangGraph, LangChain, OpenAI GPT-4o, ChromaDB, FAISS, FastAPI, PostgreSQL, Docker, GitHub, n8n

Key Responsibilities

- Developed multi-agent workflows for campaign planning and automation.
- Built Python tools integrating CRM and digital marketing APIs.
- Developed RAG pipelines for SEO and campaign knowledge retrieval.
- Implemented semantic search using ChromaDB and FAISS.
- Designed prompt templates for AI-driven marketing content generation.
- Integrated marketing analytics and reporting APIs.
- Built evaluation scripts for AI-generated campaign recommendations.
- Containerized AI applications using Docker.

Achievements

- Reduced campaign planning time by 74%.
- Automated 84% of repetitive marketing operations.
- Improved lead qualification accuracy by 38%.
- Increased campaign execution productivity by 41%.
- Reduced manual reporting effort by 69%.

Flagship Project 2: Marketing Analytics Copilot

Client: Enterprise Marketing Technology Company

Domain: Marketing Technology (MarTech)

Duration: July 2025 – Present

Role: Agentic AI Engineer

Business Challenge

Marketing teams struggled to consolidate campaign data, customer engagement metrics, website analytics, and advertising performance from multiple platforms. Manual analysis delayed optimization decisions and reduced campaign effectiveness.

Architecture & Technical Implementation

Developed an Agentic AI-powered marketing analytics platform using LangGraph, CrewAI, and LangChain to automate campaign insights and performance analysis. Built specialized AI agents for Marketing Analytics, Customer Engagement Analysis, Campaign Optimization, Performance Reporting, Budget Recommendations, and Customer Journey Analysis. Implemented Retrieval-Augmented Generation (RAG) pipelines over historical campaign data, marketing documentation, KPI dashboards, and customer engagement reports using ChromaDB and FAISS. Integrated AI agents with Google Analytics, CRM platforms, advertising APIs, and BI dashboards through FastAPI while implementing conversation memory, Human-in-the-Loop approval workflows, and automated evaluation pipelines.

Tech Stack: Python, LangGraph, CrewAI, LangChain, Claude 3.5 Sonnet, ChromaDB, FAISS, FastAPI, PostgreSQL, Docker, GitHub Actions, Power Automate

Key Responsibilities

- Developed AI agents for campaign analytics and optimization.
- Built reusable Python tools for marketing platform integrations.

- Developed RAG pipelines for marketing documentation and KPI retrieval.
- Implemented structured prompt engineering for marketing insights.
- Integrated AI agents with analytics dashboards and CRM systems.
- Developed conversation memory using ChromaDB.
- Built automated evaluation frameworks using RAGAS.
- Implemented Human-in-the-Loop workflows for campaign approval.

Achievements

- Reduced campaign analysis time by 71%.
- Improved marketing insight accuracy by 37%.
- Automated 79% of reporting activities.
- Increased customer engagement analysis efficiency by 36%.
- Reduced manual marketing operations by 63%.

Education

B.Tech. Artificial Intelligence and Data Science *Graduated: April 2026*

Nehru Institute of Engineering and Technology · Coimbatore, Tamil Nadu

CGPA: 8.2 / 10.0